

PalmSource®

Conference and Expo



Tools Overview and Roadmap

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An overview of the Palm development tools and how Palm is moving into the future with ARM devices and native development.



Agenda Slide

- Mission Statement
- Our Commitment
- Past, Present and Future
- Roles and Responsibilities
- The Development Community



Agenda Slide

- Key Initiatives
- The Tools



Mission Statement

To provide compelling, state of art development tools to internal and external developers of Palm OS® related projects.



Our Commitment

- Powerful development tools
- High quality development products
- Support of top three development platforms
- Tool of choice, on the platform of choice
- Delivery and accountability
- Customer satisfaction
- Tools for internal and external development, including Palm OS licensees



The Past

- In 1996, there was no Tools group; In 1997 there was a team of one
- SDKs and CDKs were hand-built, and never released with OS
- Extensively utilized OS and HotSync® resources
- No consistency, API evaluation or sanitization
- One location for developers



The Present

- Tools group consists of four strong sub-teams
- SDKs and CDKs are automated and released simultaneously with OS
- Team of 24
- Tools team has staffed up and has removed *reliance* on OS and HotSync teams
- APIs are consistent, evaluated and sanitized
- Development in Santa Clara, Austin, Seattle area, and Nevada City



The Future

- Tools releases not tied to platform releases
- More focus on wireless and enterprise tools



Roles and Responsibilities

- John Dance
  - Java development and specifications
  - **John.Dance@corp.palm.com**
- Eric Omelian
  - Conduit Development Kit
  - **Kelly.mccaw@corp.palm.com**
- Phillip Shoemaker
  - Tools strategy and core tools
  - **Phil.Shoemaker@corp.palm.com**
- Robert Sirota
  - Software Development Kit and Licensee Tools
  - **Robert.Sirota@corp.palm.com**



The Palm Development Community

- Windows *Significantly* Leads.
  - 82% develop on Windows
  - 11% develop on Macintosh
  - 5% develop on Linux
  - 2% develop on other (including Amiga)
- What else?
  - Non-Metrowerks
  - C++ Frameworks
  - Where are all the chicks?
  - Java, Java, Java
  - Free development environments



Key Initiatives

- Tool of choice, platform of choice
- Short learning curve
- Inexpensive tools
- Tools for all
- Empowering our partners



Key Initiatives, Part Two

- Three tiered OS development
- Tools for the user



Tool of choice, platform of choice

- Using Windows?
  - We plug in to Visual Studio!
  - Conduits today, more tomorrow
- Using Macintosh?
  - We plug in to CodeWarrior!

- Use the tool you want to use.



Short Learning Curve

- To lessen the learning curve, our tools will…
  - integrate *fully* into your chosen environment
  - utilize the rapid application development functionality found in your chosen environment
    - AppWizards
    - ClassWizards
  - include context help, including API calls
  - include code snippets and samples
  - include the Palm OS built-in apps with projects for the various supported environments



Inexpensive Tools

- No expensive tools
- No extraneous environments
  - *Tool of choice for platform of choice*
- Free?



Tools for all

- Application development for all Palm OS® ARM devices
- Tools that generate quality, portable ARM code for Palm OS devices
- No special device optimizations/customizations
  - That's where our partners pitch in

Empowering our partners

- Providing our partners with kits and documentation to…
  - Document how partners integrate with the Palm OS Emulator
  - Palm Universal Debugger integration…
    - How do licensees attach to a new protocol?
    - How do the silicon partners attach to a new reference board?
    - How do tools vendors modify their debugger to support the new solution?
  - Palm Compiler/Linker/Assembler/Librarian
    - Code generation
    - PRC creation
    - Palm runtime format

Three tiered development

- Develop most of your application in the Simulator
  - Gives you quick debugging and running
  - Allows you to use platform specific tools
  - Still a simulator… not 100% compatible
- Use the Palm OS Emulator
  - Runs on a variety of platforms
  - Conducts many checks on code safety and ‘behavior’
  - Hardware emulator… close, but not perfect
- Debug on the device
  - Slower, but finds the problems

Tools for the user

"Tools that do not hurt the end user experience."

- Java
  - MIDp
    - Provides a lowest common denominator user interface experience
    - Covers pagers, cell phones and handheld computers
  - PDAp
    - Provides a user interface specific to handheld computers, giving users a consistent feel on the Palm OS platform



The Tools

- Conduit Development Kit
- Software Development Kit
- Compilers and Linkers
- Palm Universal Debugger
- Partner Tools



The Tools

- Palm OS Emulator
- The Palm Simulator
- Resources
- Java
- Automated Testing Environment



Conduit Development Kit

- Conduits are crucial
- Wizards for ease of development
- COM-enabled conduits
  - Visual Basic
  - Delphi
  - JavaScript
- Generic Conduit as a C++ cross platform strategy
- Investigating Java as a cross platform conduit mechanism



Software Development Kit

- Headers and libraries
- Sample applications
- Application Frameworks
- Licensee support
- ResEdit support
- Tutorial
- Documentation
- Resource Editor



Compilers and Linkers

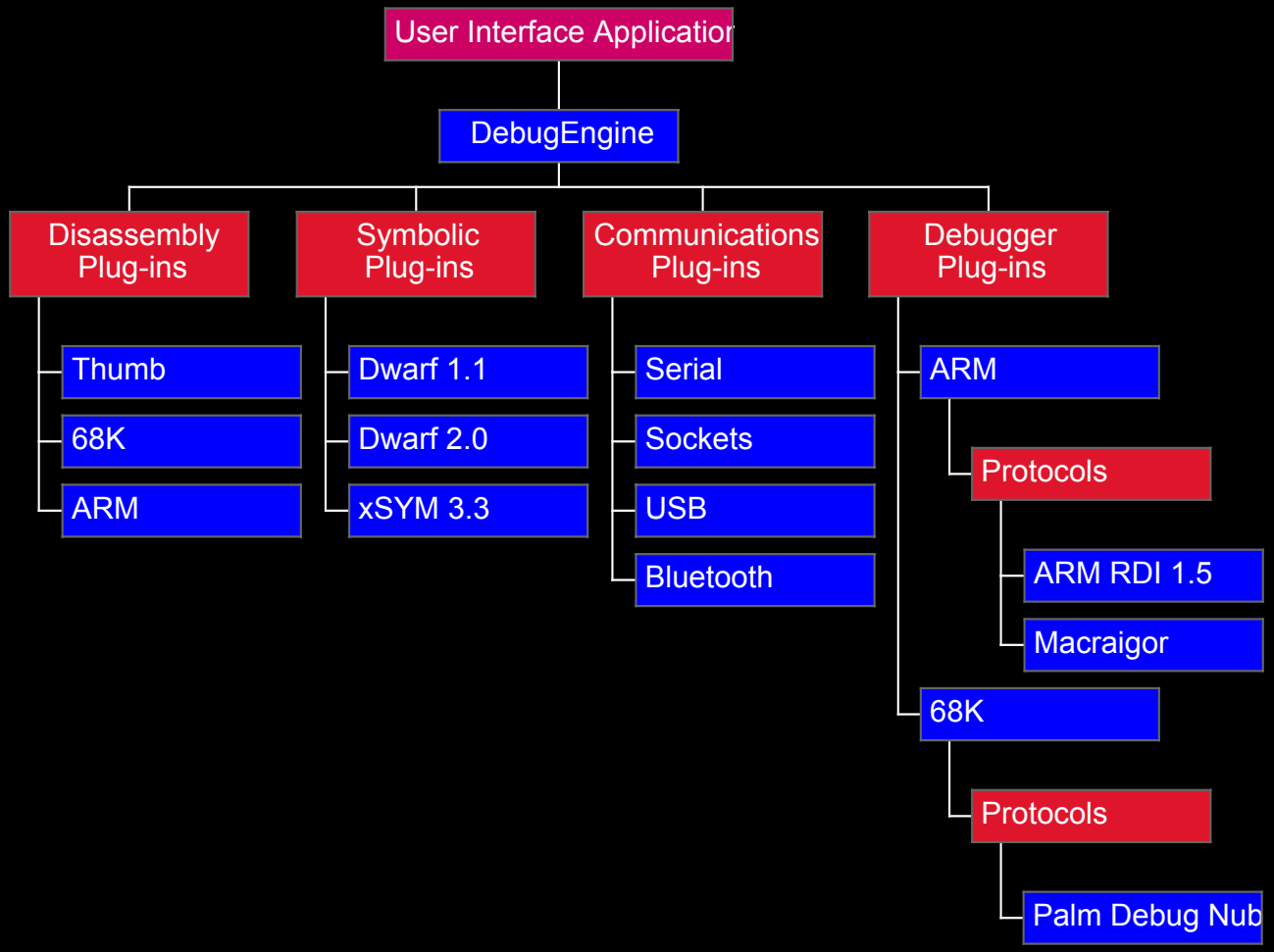
- Targeting ARM v4
- EDG Front-end
- IR Optimizer and back end technologies acquired from Apogee Software, Inc.
- Tools will be inexpensive (if not free)
- Stand alone technologies on Macintosh and Windows
- Plug-in technologies for Macintosh and Windows



Palm Universal Debugger

- Design Goals
  - Full featured debugger
  - Expandable
  - Flexible
  - Re-usable code
  - Cross platform code
  - Provide good user experience
- Supports
  - ARM
  - 68k
  - xScale

Palm Universal Debugger





Palm Universal Debugger

Demo



Partner Tools

- ROM Building Tools
- Flash Tools
- Debugger SDK
- File Format Documents
- Palm OS Emulator Training

Palm OS Emulator

- Past
  - Stricter checking for Palm OS 5 compatibility
    - Window, form, and form object access
    - Proscribed function calls
- Present
  - Event minimization
  - Memory Leak Reporting
- Future
  - ARM Support
  - Feature parity with PalmSim



The Palm Simulator

- Native code
- Fast execution
- Fast debugging
- Native tools
  - Visual Studio
  - Purify, HeapAgent, etc.



Resources

Resource format and compiler

- New text based resource description (XML-based) (XRD Files)
- Cross-platform support
- Command line support for all platforms (PalmRC)
- Plug-in support for top development environments
- Multiple language support



Resources

Resource Editor

- Constructor is getting phased out
- Support for variety of platforms:
  - Windows
  - Macintosh
  - Linux
- Standalone and plug-in technologies
- Multiple language support



Java

- Enabling millions of developers
- Enabling all Java implementations
- Evangelizing the PDA Profile (PDAp)
  - Maintaining the Palm user experience
  - Minimum set of common functionality
  - Application portability across devices
  - Complement Palm OS functionality
  - Open licensing as part of JCP

Java

What is the PDA Profile?

- A new J2ME/CLDC Profile (JSR 75) that fits the needs of Personal Digital Assistants:
  - Runs in a limited processing environment
  - Supports a sophisticated UI widget set
  - Accesses standard Personal Information Management (PIM) features
  - Support connectivity to other devices
- Palm OS Subsidiary is Specification Lead
  - Brad Jarvinen, Brad.Jarvinen@corp.palm.com
- Many key industry companies on expert group (Sun, OTI, Motorola, RIM, Sharp, Nokia to name a few)
- Following the Java Community Process (www.jcp.org)



Automated Testing Environment

- Record and playback of user interactions within the Palm OS
  - On a device
  - On the Palm OS Emulator
- Can modify scripts with an editor
- Allows for capture of full regression tests
- Can work on single application or multiple applications



Q&A



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